

CM4125 Topic 3 Lab



Setting

- 6 players
 - 3 crewmates
 - 3 impostors
- Admin (me)
 - Assign roles
 - Keep time
 - Provide visualisations
 - “Kill” players
 - Keep track of the crew’s suggestions
- The rest of you will be public/voters



Rules

1. The admin will provide a “bad” visualisation to everyone
2. All players (and the public!) have ~5 minutes to see it and think about how to “improve” the visualisation
 1. Crewmates want to make it better
 2. Impostors want to (secretly) make it worse, or not improve it at all
3. Each player will give the admin one suggestion
4. Once all suggestions have been given to the admin, an emergency meeting is called!
5. Players have ~5 minutes to debate in public who the impostor is/are
6. After that, the public votes for someone to be sent off
 1. The person with the most votes gets “killed” (out of the game)
7. Afterwards, a new visualisation will be provided, and the process starts again



Consider that...

- All players must be actively participating during the emergency meeting
- Play your strategy right and try to guess who is in your team
- If you feel we have reached a dead end with an example, then the admin can provide a new one



Let's Play!

Vis 1

♦ AI Insights

Coffee is boosting your grades

Weekly coffees vs quiz score · Semester 1 · 11 weeks analysed

$r = 0.97$

Very strong correlation

+2.1 marks

per extra coffee a week

68%

Average quiz score

95%

Projected exam score (W15)



♦ Key finding

Your data shows a clear causal link between caffeine and academic performance. Each additional coffee per week drives a 2.1-mark improvement in quiz results.

Recommendation: increase to 28 coffees per week to reach 100% by the final exam.

Confidence: High ●●●●○

Source: Journal of Applied Caffeine Studies (2024); your weekly quiz results, Weeks 1–11.

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Answer key: what the machine got wrong

Rule 1 · Bar axis starts at 4, not 0
W1 (6 coffees) is 30% of W11 (20), but its bar is drawn at about an eighth of the height.

Rule 3 · Two y-axes
The right-hand scale is set so the score line sits on the bar tops. Rescale either axis and the 'match' goes.

Machine #1 · Week 5 was never recorded
Reading week has no data. The bar and the point are interpolated, and nothing on the chart says so.

Rule 7 · Extrapolation
A straight line through 11 weeks is projected to 95% in the exam. The same line passes 100% before week 17.

Rule 8 / Machine #3 · Correlation sold as cause
Coffee and scores both rise as exams get closer. Study time is the obvious confounder. 'Confidence: High' is decoration.

Machine #4 · Mean of percentages
The 68% averages weekly percentages from quizzes of 10 to 50 questions, including the invented week. Real: $188/260 = 72\%$.

Machine #5 · Source doesn't exist
The 'Journal of Applied Caffeine Studies' was invented for this exercise. Always click the source.

Rule 8 · A slope treated as a dose
'+2.1 marks per coffee' leads to the recommendation of 28 coffees a week. A regression slope is not a prescription.

The other examples

2. <https://www.matillion.com/wp-content/uploads/2015/01/bad-dashboard-examples-1.png>
3. <https://www.matillion.com/wp-content/uploads/2015/01/bad-dashboard-examples-3.png>
4. <https://wp.klientboost.com/wp-content/uploads/2013/05/91.png>
5. <https://wp.klientboost.com/wp-content/uploads/2013/05/5.jpg>
6. <https://cdn-wordpress-info.futurelearn.com/wp-content/uploads/37e0c522-1345-47e6-9a29-0030ff18fd6a-754x451.jpg.webp>